HASSAN AHMAD

AI / ML Engineer | Python · Computer Vision · Chatbot Development · API Integration
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RESEARCH PROFILE

B.Sc. Software Engineering graduate (CGPA 3.54/4.0) with a research record in applied machine learning and deep learning: a final-year research project and a co-authored paper, both with manuscripts under journal preparation. Experienced in time-series forecasting with external variables, model evaluation and reproducible Python workflows, built across four industry internships. Applying for the project "Integrated AI-Driven Carbon Intelligence Analytics for Sustainable Urban Campuses", with a strong interest in applying AI to sustainable energy systems.

RESEARCH EXPERIENCE & PUBLICATIONS

Multimodal Driver Alertness Classification 2025 – 2026
University of Malakand · Final Year Research Project

- Designed and implemented a deep learning pipeline (Python, PyTorch/TensorFlow, OpenCV) for multimodal classification that remains robust when input data is missing.
- Authored the full research manuscript for JKSUCIS Journal: "Deep Learning-Based Multimodal Affective Driving-State Risk Classification With Missing-Modality Robustness" (2026, ongoing).

Intelligent Heart Murmur Detection 2025 – 2026
University of Malakand · Research Project (co-author)

- Co-developed a signal-processing and deep learning system: CWT scalograms with an attention-based CNN benchmarked against nine transfer-learning models; 99.53% accuracy on the CirCor dataset.
- Co-authored "Intelligent Heart Murmur Detection Using PCG Signals and Attention-Based CNNs" (2026, ongoing).

TECHNICAL EXPERIENCE

Machine Learning Engineer (Intern) Oct 2025 – Mar 2026
ITSOLERA Pvt. Ltd. · Islamabad, Pakistan

- Built a **Time-Series Demand Forecasting System** using external variables (such as weather and economic indicators), comparing ARIMA, SARIMA, Prophet, Random Forest and XGBoost with MAE, RMSE and MAPE and rolling-window validation.
- Created an **Adaptive Synthetic Data Augmentation Toolkit** (GANs, VAEs, SDV) that analyses model errors and generates targeted synthetic data to improve performance on under-represented classes.
- Built an image classification system (HOG features; SVM, KNN, Random Forest) and a content-based recommendation system; version-controlled with Git/GitHub.

Generative AI Intern (Remote)
Neuro Five Solutions Aug 2026 – Sep 2026

- Built LLM applications including a RAG question-answering system, a multi-agent workflow, validated JSON outputs and an AI workflow automation, plus a full-stack AI capstone.

Machine Learning Engineer (Intern, Remote) Jan 2025 – Mar 2025
Arch Technologies · Cognifyz Technologies

- Delivered end-to-end ML projects (regression, NLP classification, recommendation, geospatial analysis) covering preprocessing, feature engineering, training, evaluation and documentation of findings.

TECHNICAL SKILLS

AI / ML	Machine learning, Deep learning, Lang Graph, Lang Chain, Py Torch, RAG, Liama 3.1, Hugging Face.	**Backend**	Python, Fast API, REST APs
Programming	Python, C++, SQL, Algorithms, Data Structures	**Data**	Pandas, NumPy, SQL, Time-Series Analysis, Feature Engineering
Languages	English (C1), Urdu (Native), Pashto (Native)	**Tools & DevOps**	Git, GitHub, Jupyter Notebook, VS Code, Docker (basic)

EDUCATION

B.Sc. Software Engineering · CGPA 3.54 / 4.0 Sep 2022 – Jun 2026
University of Malakand, Pakistan

CERTIFICATIONS & MEMBERSHIPS

- Machine Learning Specialization — Stanford University (Coursera)
- Deep Learning — Deep Learning.AI
- IEEE Student Member, University of Malakand Chapter

REFERENCES

Dr. Khalil Ullah
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